Numerical Expressions & Factors: Prime Factorisation & GCF

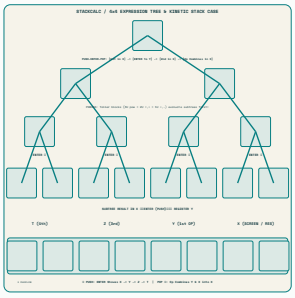
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.6.NS.B.4

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Tree Board

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Decompose composite numbers into factor branches on the tree board.

Sketch Model on Paper:
Draw a factor tree breaking 36 into 2 × 2 × 3 × 3.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	36	X: 36	Composite 36 into X
2.	ENTER	Y: 36 | X: 36	Stack lift
3.	2	Y: 36 | X: 2	Test smallest prime
4.	÷	X: 18	Quotient: 36 ÷ 2 = 1

EXPECTED LCD X: 18 (2 × 18)

1 **Divide 48 by prime factor 2 on stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Divide 75 by prime factor 3 on stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Find GCF of 24 and 36: test factor 12.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Find LCM of 4 and 6 on the stack: (4×6) ÷ 2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 1

Numerical Expressions & Factors: Prime Factorisation & GCF • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: 4x4 Expression Tree Case & Factor Tokens

CAD Blueprint: tree_board

Target Device: SC-6 / SC-32 Bridge

CAD / STL Source File:

Education/build/scad/tree_board.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Expression Tray & White Tokens (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Divide 48 by prime factor 2 on stack.** Standard: CCSS.MATH.6.NS.B.4	24	[4][8] [ENTER] [2] [÷]
2	**Divide 75 by prime factor 3 on stack.** Standard: CCSS.MATH.6.NS.B.4	25	[7][5] [ENTER] [3] [÷]
3	**Find GCF of 24 and 36: test factor 12.** Standard: CCSS.MATH.6.NS.B.4	3 (Zero rem)	[3][6] [ENTER] [1][2] [÷]
4	**Find LCM of 4 and 6 on the stack: (4×6) ÷ 2.** Standard: CCSS.MATH.6.NS.B.4	12	[4] [ENTER] [6] [×] [2] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for confusing factors (divisors) with multiples.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.